Si risolvano le seguenti equazioni e disequazioni:

1. $\dfrac{x-1}{x^2-5x+6} \leq 0;$
2. $\dfrac{x^3-2x+1}{x+2} \geq 0;$
3. $\dfrac{x^3-3x^2+2x}{x-4} < 0;$
4. $|x+1|+2x-3=2;$
5. $\left|\dfrac{x^2-2}{x}\right| \leq 1;$
6. $\left|\dfrac{x^2-4x+3}{x-5}\right| < 2;$
7. $\dfrac{|x^2-4|+2x}{x+1} \leq 0;$
8. $\sqrt{x^2-9} = x-4;$
9. $\sqrt{x^2-4x} \leq x+3;$
10. $\sqrt{x^2-5x+6} \geq x+4;$
11. $\sqrt[3]{\dfrac{x^2-2}{x}} = 1;$
12. $\sqrt{|x^2-1|-2} = x+2;$
13. $\sqrt{x^2-1}-x \geq 0;$
14. $\sqrt{x^2-3x+2} \leq x-3;$
15. $\sqrt{x^4-x^2} \geq x^2-1;$
16. $\left(\dfrac{1}{4}\right)^x < 3;$
17. $\left(\dfrac{1}{4}\right)^x - \left(\dfrac{1}{2}\right)^x + 2 > 0;$
18. $9^x - 5\dot{3}^x + 6 \leq 0;$

19. $2^{-x} + 2^x - 3 \geq 0$;

20. $\log^2_{1/3} x - 2\log_{1/3} x + 3 \leq 0$;

21. $\log\left(\dfrac{x^2-3}{x+1}\right) \geq 0$;

22. $\log_{1/2}\left(\dfrac{x^2-x-2}{x^2-3}\right) \geq 0$;

23. $\log_{\frac{1}{3}}(x^2 - x) \leq 1$;

24. $\log_3(x^2 - 1) > 2$;

25. $\log^2_{\frac{1}{2}} x - 4\log_{\frac{1}{2}} x > 0$;

Si risolvano i seguenti sistemi di disequazioni:

$$\begin{cases} \dfrac{x+3}{2x-1} \leq 2 \\ \dfrac{x^2}{x^2+x} > 1 \end{cases}$$

$$\begin{cases} \sqrt{x^2+2x} \leq 2 \\ \dfrac{x+1}{x-2} \leq 3 \end{cases}$$

$$\begin{cases} \dfrac{x^2-3x}{x+1} \leq 0 \\ \dfrac{3x-1}{x} \geq 1. \\ \dfrac{x^2-4x}{x+2} \leq 0. \end{cases}$$

$$\begin{cases} \log\left(\dfrac{x+2}{x^2}\right) \geq 0 \\ \\ e^{2x} - 3e^x + 2 \leq 0. \end{cases}$$

Si determini il dominio delle seguenti funzioni:

1. $f(x) = \dfrac{x^2 - 3x}{x+1} + \sqrt{x-5}$;
2. $f(x) = \log_3\left(\dfrac{x-3}{x^2-4}\right)$;
3. $f(x) = \sqrt[3]{\dfrac{x^2+4}{x^3-4}} + \dfrac{x}{x^2-x}$;
4. $f(x) = \dfrac{3}{\sin x} + \sqrt{\dfrac{1}{x^2-1}}$;
5. $f(x) = \arccos(x^2 - 2x)$.

Si determinino il dominio e la positivitá delle seguenti funzioni:

1. $f(x) = \log_{\frac{1}{3}}\left(\dfrac{x-3}{x^2-4}\right)$;
2. $f(x) = \sqrt{x^2-x} - 2x$;
3. $f(x) = \arcsin(x^2 - x)$;
4. $f(x) = \dfrac{x^2-1}{x+2} - 1$;
5. $f(x) = 2^{\frac{x-1}{x+1}} - 1$.